Computer Science — Class 6

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 2 — Components of Computer System — Simplified Exam Notes for School-Based Assessment (SBA)

TOPIC 1: BASIC COMPONENTS OF A COMPUTER:

A computer is a machine that follows the instructions given by its user. A computer system is made up of five basic components, and each one has its own job to do.

Long Question: Discuss the basic components of a computer system.

A computer system has five basic components, and each one works as follows:

Input Devices: Input devices are those devices that are used to enter data into the computer. **Example:** keyboard, mouse.

Output Devices: Output devices are those devices that are used to show the result to the user. **Example:** monitor, printer.

Memory Unit: The memory unit is the part of the computer that stores data and instructions so that they can be used later.

Control Unit: The control unit is the part of the computer that manages and directs the whole working of the computer.

Arithmetic and Logical Unit (ALU): The ALU is the part of the computer that carries out all the maths and comparisons that the computer performs.

CPU (Central Processing Unit): The CPU is the part of the computer formed when the memory unit, the control unit and the ALU work together, and it is also known as the brain of computer.

TOPIC 2: INPUT DEVICES AND SENSORS:

An input device is used to give data to the computer. This data can be in the form of numbers, letters, or images, and the computer changes it into useful information.

Long Question: Discuss any five input devices in detail.

The commonly used input devices and their details are as follows:

Keyboard: The keyboard is the most common input device, and it is used to type data into the computer. **Example:** it has 84 keys, 101 or 102 keys, and newer keyboards come with 104 or 108 keys.

Mouse: The mouse is a pointing device that is used to move and control the cursor. **Example:** it has three buttons, namely left, right and scroll.

Joystick: The joystick is also a pointing device that is mostly used for playing games.

Light Pen: The light pen is a pen-like pointing device that is used to select items or to draw on the screen.

Scanner: The scanner is a device that works like a photocopy machine and changes a paper document into digital form. **Example:** a paper document, called a hardcopy, becomes a digital file, called a softcopy.

Microphone (Mic): The microphone is a device that is used to input sound into the computer.

OCR (Optical Character Reader): The OCR is a device that reads printed text and turns it into editable text.

Bar Code Reader: The bar code reader is a device that reads bar codes and sends the value to the computer. **Example:** commonly seen in shopping malls.

2.3 Sensors: Sensors are input devices that record data about the physical environment around them. They cannot make decisions on their own; they only send data to the computer. **Example:** the magnetic field sensor, the motion detector sensor, the sound sensor, the humidity sensor, the pressure sensor, and the light sensor.

TOPIC 3: OUTPUT DEVICES: Output devices show the result of processing to the user. **Example:** monitor, printer, speaker.

Long Question: Discuss output devices. Describe the three types of monitors.

A monitor is also called a Video Display Terminal (VDT) or a Video Display Unit (VDU). It is an output device that shows images, text, video and graphics, and looks like a TV, but has a much higher resolution. There are three types of monitors:

CRT (Cathode-Ray Tube): A CRT monitor is a type of monitor that is old and heavy, and it uses electron beams to create the picture on the screen.

Flat-Panel Display: A flat-panel display is a type of monitor that is thin and light with a higher resolution. **Example:** emissive types such as plasma and LED, or non-emissive types such as LCD.

SMD LED Screen: An SMD LED screen is a type of monitor that uses LEDs mounted on circuit boards; SMD stands for Surface Mounted Device.

Other common output devices are as follows:

Multimedia Projector: A multimedia projector is a high-resolution projector that shows text, images, video and audio on a big screen; it is also called a data projector.

Graphic Plotter: A graphic plotter is a device that prints vector graphics using a pen, a pencil or a marker, and it is mainly used for computer-aided design.

Printer: A printer is a device that prints text and images on paper. **Example:** the inkjet printer and the laser printer.

Speaker: A speaker is a device that generates sound, and the signal for this sound comes from the computer's sound card.

TOPIC 4: CENTRAL PROCESSING UNIT (CPU), MEMORY AND STORAGE:

Long Question: Write a note on CPU. Discuss its three main parts. *[Exercise]*

The Central Processing Unit (CPU) is the core of a computer system, and it has three main parts:

Memory Unit: The memory unit is the part of the CPU that stores data. **Example:** it has two types, namely primary memory and secondary storage.

Control Unit: The control unit is the part of the CPU that manages the whole working of the computer by collecting the input, sending it for processing, and then giving the output to the user.

Arithmetic and Logical Unit (ALU): The ALU is the part of the CPU that performs addition, subtraction, multiplication, division, and comparisons such as less than, greater than, and equal to.

Primary Memory: RAM and ROM:

RAM (Random Access Memory): RAM is a type of primary memory that is also called short-term or temporary memory. It is volatile, which means the data stored in it is lost when the power goes off, and it stores the data that is currently being processed.

ROM (Read Only Memory): ROM is a type of primary memory that is also called long-term or permanent memory. It is non-volatile, which means the data stored in it stays even when the power goes off, and it holds the instructions needed to start, or boot up, the computer.

Secondary Storage: Secondary storage is the part of the computer that keeps data permanently, and it is cheaper and larger than primary memory, though the CPU cannot access it directly.

HDD (Hard Disk Drive): The HDD is a secondary storage device that stores data on magnetic plates. **Example:** it has a large capacity and is faster than a DVD, but it has moving parts that can be damaged, and it uses more power and makes more noise.

SSD (Solid State Drive): The SSD is a secondary storage device that is non-volatile and uses flash memory, which makes it much faster than an HDD. **Example:** it is durable, compact and quiet, but it is more expensive and has a shorter life.

USB Flash Memory: USB flash memory is a secondary storage device that is non-volatile and works on Plug and Play, so no driver is needed. **Example:** it can store up to 1 terabyte of data; it is cheap, fast and small, but it can spread malware and is easy to lose.

Computer Memory Measurement Units:

Symbol	Full Form	Quantity
1 Bit	Binary Digit	1 cell, binary 0 or 1
4 Bits	Nibble	1/2 Byte
8 Bits	Byte	1 Byte
1024 Byte	Kilobyte (KB)	1 KB
1024 KB	Megabyte (MB)	1 MB
1024 MB	Gigabyte (GB)	1 GB
1024 GB	Terabyte (TB)	1 TB
1024 TB	Petabyte (PB)	1 Petabyte
1024 PB	Hexabyte	1 Hexabyte
1024 Hexabyte	Zeetabyte	1 Zeetabyte

TOPIC 5: WORKING OF COMPUTER SYSTEM:

Long Question: How does a computer work? Explain its four main operations. *[Exercise]*

A computer works through four main operations:

Input: Input is the data before processing, and it comes from input devices. **Example:** keyboard, mouse, or a sensor.

Processing: Processing is the operation that changes the input into output, and this work is done by the CPU.

Storage: Storage is the operation in which the computer saves data. **Example:** RAM stores it for a short time, while the hard drive stores it for long-term use.

Output: Output is the final result of processing. **Example:** shown through a monitor, a printer, or a projector.

Stages of Data Processing: Data goes through six stages before it becomes useful information, in this order: data collection, data preparation, data input, data processing, data interpretation, and data storage.

SHORT QUESTIONS — CHAPTER 2

1. What are input devices? Enlist any four. ***[Exercise]***
Devices through which a user feeds data into a computer are called input devices. Four such devices: keyboard, mouse, scanner, microphone.

2. Define Input. Also, list any three input devices. ***[Paper — SBA Mid-Term]***
Whatever a user feeds into the computer for processing is called input. Three input devices: keyboard, mouse, scanner.

3. Write down the type of the scanner and the mouse. ***[Paper — SBA Mid-Term]***
Both scanner and mouse belong to the input device category.

4. Write a short note on keyboard. ***[Exercise]***
The keyboard lets a user type letters, numbers and symbols straight into the computer. Older keyboards had 84 keys, while most keyboards today use 101, 102, 104, or 108 keys.

5. Differentiate between mouse and joystick. ***[Exercise]***
A mouse is held in the palm and has three buttons to move the cursor, while a joystick is a stick-shaped device operated mainly for gaming.

6. How does a light pen work? ***[Exercise]***
A light pen has a photocell. When its tip touches the screen and the button is pressed, it sends the screen location to the CPU.

7. What is scanner? Why do we use it? ***[Exercise]***
A scanner captures whatever is printed on paper and saves it as a computer file, much like a photocopier captures a page, so the content can be edited or reused on a computer.

8. How does a barcode reader work? ***[Exercise]***
A barcode reader shines light on the black-and-white bar pattern printed on a product, reads the pattern, and passes the matching code number to the computer.

9. What are sensors? Where do we use them? ***[Exercise]***
A sensor is a device that senses changes around it, such as light, heat or motion, and passes this information to the computer. Sensors are commonly built into alarms and automatic doors.

10. Write a note on monitors. Enlist its types. ***[Exercise + Paper]***
A monitor displays whatever the computer processes, and is also known as a VDT or a VDU. Based on the screen technology used, monitors fall into three kinds: CRT, Flat-Panel, and SMD LED.

11. Differentiate between Flat panel and SMD monitors. ***[Exercise]***
A flat-panel monitor is a slim screen built using plasma, LED or LCD technology, whereas an SMD screen specifically means LEDs fixed onto a circuit board (Surface Mounted Device).

12. Which of the following is not an input device? ***[Exercise concept]***
Speaker is not an input device. It is an output device.

13. Differentiate between primary memory and secondary storage. ***[Exercise]***
Primary memory, such as RAM and ROM, is fast but limited and costly, while secondary storage, such as a hard disk or SSD, holds far more data at a lower cost.

14. Differentiate between RAM and ROM. ***[Exercise]***
RAM loses its content once the computer is switched off, so it only holds data while the computer is running; ROM keeps its content permanently and carries the instructions the computer needs when it first starts.

15. How is SSD better than HDD? ***[Exercise]***
An SSD has no moving parts, so it reads and writes data much more quickly, runs silently, and consumes less power compared to a traditional HDD.

16. Write the function of the control unit. ***[Exercise]***

The control unit acts like a traffic controller inside the computer: it takes in the input, routes it for processing, and then passes the result on as output.

17. Why is CPU called the brain of computer? ***[Book concept]***

CPU controls all parts of the computer. No action can happen without its permission.

18. Can a computer work without secondary storage? Justify. ***[Exercise]***

A computer can process data using only RAM and ROM. But it cannot save data permanently without secondary storage.

19. How does a computer work? Write the steps to convert data into information. ***[Exercise]***

A computer takes input, processes it, stores it, and gives output. This turns data into information.

20. Write the function of ROM. ***[Exercise]***

When a computer is switched on, ROM supplies the built-in startup instructions it needs before anything else loads.

21. Write the function of RAM. ***[Exercise]***

RAM temporarily holds whatever the CPU is actively working on at a given moment.

22. Write the function of ALU. ***[Exercise]***

The ALU is the calculator of the CPU: it carries out arithmetic (add, subtract, multiply, divide) and compares values (less than, greater than, equal to).

23. Write the function of HDD. ***[Exercise]***

An HDD keeps the computer's data saved for the long run by writing it magnetically onto spinning metal plates inside the drive.

24. Write the function of SSD. ***[Exercise]***

An SSD saves large volumes of data on memory chips instead of spinning disks, which is why it works faster and more reliably than older drives.

25. What is the function of a microphone? ***[Book concept]***

A microphone picks up spoken or recorded sound from the surroundings and converts it into a digital file inside the computer.

MULTIPLE CHOICE QUESTIONS — CHAPTER 2

1. Which of the following is <u>not</u> an input device? ***[Exercise MCQ 1]***

a) keyboard **b) speaker** ✓ c) mouse d) scanner

2. Mouse is mainly used to:

a) type text **b) move cursor** ✓ c) print document d) scan image

3. In light pen, the sensing element is called: ***[Exercise MCQ 2]***

a) photocell ✓ b) cell c) light cell d) none

4. Input device used to transfer hardcopy into computer: ***[Exercise MCQ 3]***

a) mouse b) joystick **c) scanner** ✓ d) keyboard

5. Device used to input sound into the computer: ***[Exercise MCQ 4]***

a) joystick b) speaker c) light pen **d) microphone** ✓

6. OCR stands for: ***[Exercise MCQ 5]***

a) optical Correct Reading **b) optical Character Reader** ✓ c) optimal Copy Reading d) optical Character Reading

7. Which of the following is/are type of sensor? ***[Exercise MCQ 6]***

a) Sound sensor b) Pressure sensor c) Light sensor **d) all** ✓

8. Devices used to display the result of processing are called: ***[Exercise MCQ 7]***

a) output devices ✓ b) input devices c) storage devices d) processing devices

9. Which of the following is/are type of monitor? ***[Exercise MCQ 8]***

a) LCD b) TFT c) SMD **d) all** ✓

10. Joystick is mainly used for:

a) typing **b) playing games** ✓ c) printing d) scanning

11. A mouse usually has how many buttons?

a) one b) two **c) three** ✓ d) four

12. A keyboard with 104 or 108 keys is an example of a/an:

a) output device **b) input device** ✓ c) storage device d) sensor

13. Bar Code Reader is used to read:

a) sound **b) bar coded data** ✓ c) light d) pressure

14. A printed copy of a document is also called:

a) softcopy **b) hardcopy** ✓ c) scancopy d) none

15. A scanner changes a paper document into:

a) hardcopy **b) digital form** ✓ c) sound d) bar code

16. Which device is <u>not</u> used for pointing on the screen?

a) mouse b) joystick c) light pen **d) microphone** ✓

17. A monitor is also known as:

a) VDU ✓ b) CPU c) ALU d) ROM

18. A CRT monitor creates the picture using:

a) flash memory **b) electron beams** ✓ c) solid state chips d) bar codes

19. Which of the following is/are <u>not</u> part of CPU? ***[Exercise MCQ 9]***

a) ALU b) MU **c) PU** ✓ d) CU

20. All mathematical operations in a computer are performed by: ***[Exercise MCQ 10]***

a) memory unit b) control unit **c) arithmetic unit** ✓ d) storage unit

21. Which unit manages the entire working of the computer?

a) ALU **b) Control Unit** ✓ c) RAM d) ROM

22. RAM stands for:

a) Random Access Memory ✓ b) Read Access Memory c) Random Allowed Memory d) Read Allowed Memory

23. ROM stands for:

a) Read Only Memory ✓ b) Random Only Memory c) Read Open Memory d) Random Open Memory

24. Which memory is volatile (temporary)?

a) ROM **b) RAM** ✓ c) HDD d) SSD

25. Which memory holds the boot-up instructions and does not lose data when power is off?

a) RAM **b) ROM** c) SSD d) USB

26. A Hard Disk Drive stores data on:

a) flash chips **b) magnetic plates** c) paper d) light beams

27. SSD is faster than HDD because it uses:

a) magnetic plates **b) flash memory** c) paper tape d) vacuum tubes

28. USB Flash Memory is a _____ device, so no driver is needed.

a) Plug and Play b) Read Only c) Volatile d) Analog

29. 8 Bits make one:

a) Kilobyte **b) Byte** c) Megabyte d) Nibble

30. 1024 Bytes make one:

a) Bit b) Nibble **c) Kilobyte** d) Gigabyte

31. 1024 Megabytes make one:

a) Kilobyte b) Terabyte **c) Gigabyte** d) Byte

32. The first step in the working of a computer is:

a) Output b) Storage c) Processing **d) Input**

33. Changing input into output is done by:

a) Input unit **b) Processing (CPU)** c) only Output unit d) none

34. A multimedia projector is also called a:

a) scanner **b) data projector** c) joystick d) sensor

35. A plotter is mainly used to print:

a) sound files **b) vector graphics** c) bar codes d) sensor data

36. The first stage of data processing is:

a) Data Input **b) Data Collection** c) Data Storage d) Data Interpretation

Mistake Found in the Book: Figure 2.9 in the printed book is labelled **"ORC."** This should read **"OCR"** (Optical Character Reader). It is a printing mistake in the textbook. These notes use the correct term, OCR.